

6. Project Testing

Objective

The objective of testing is to verify that the application performs correctly under different input conditions and generates accurate predictions.

Test Environment

Operating System:

Windows 11

IDE:

Visual Studio Code

Programming Language:

Python

Framework:

Flask

Test Cases

Test Case 1

Input:

Valid Life Expectancy

Valid Schooling

Valid GNI

Expected Result:

Predicted HDI displayed.

Actual Result:

Prediction displayed successfully.

Status:

PASS

Test Case 2

Input:

Missing values

Expected Result:

Validation message shown.

Actual Result:

Input validation works correctly.

Status:

PASS

Test Case 3

Input:

Invalid characters

Expected Result:

Prediction should not proceed.

Actual Result:

Application handles invalid input.

Status:

PASS

Performance Testing

The prediction response is generated within a few seconds.

Result

The application passed all functional tests successfully.

Conclusion

Testing confirmed that the system performs reliably and produces accurate Human Development Index predictions.